Ghost Light

Production Manager · The Ellery Variety Theatre

15 DAYS · 49 STOPS · GRADE 11 · GHOSTLIGHT

THE OPENING CARD

The Ellery has been dark eleven years, and it opens in a fortnight with nine hundred seats sold. The safety check is on the first night, and no one has measured this house since 1958. You are the production manager, and you sign for the rig, for the seats a crowd can see from, and for the licence. In fifteen days you hand the Ellery licence file to the man who signs it. Forty-one people are on the payroll for a run that starts only if the doors open.

DAY 1 OF 15

Eleven years of dust and nine hundred seats

BEFORE THE DAY — GREET

Forty-one people, and half of them started yesterday

Forty-one people were hired separately for this fortnight and most have never worked this building, so get round as many of them as you can and put a face to each job.

PLAN CARD 156W

Monday, fourteen days out. Ruth Dowd, the box office manager, has three weeks of advance sales. The producer is quoting the last week alone as the trend for the fortnight. Alan Kettleby, the house manager, has a seat map drawn in 1911. The stage opening was made narrower in 1958, and nobody wrote down what that did to the front rows. Hettie Prosser bought row A seat 12 the morning the sale opened, and nobody can yet tell her what she will see. The licence file takes one measured number a day, and today's is the first of fifteen. Every figure in it is taken in this building, not copied off a drawing. Today you work out what the sales are really doing. You also settle which seats are in doubt, and what the shop has to build to. A number read off the flat part of a curve is a decision about a different fortnight.

BRIEFING 16W

The seat map is 1911, the proscenium is 1958, and the advance is three weeks old.

WHAT YOU DECIDE 12W

Establish what the room actually is and what the sales are doing.

1.1 Off the flat part

BOX OFFICE · A A PLACE · BALLPARK

SCENE 31W

Dowd has three weeks of advance. Week one sold 41 seats, week two 66, week three 188. Sallow, the producer, is quoting the last week as the rate for the fortnight.

GUIDE 61W

A rate of change is a difference divided by an interval, and the interval is part of the answer. Work out the average across all three weeks, then compare it with the last week on its own. Both are correct arithmetic about the same three numbers, and one of them is being used to forecast eleven days nobody has data for.

ASK 12W

Estimate the average rate of change in sales across the three weeks.

BEHIND THE BUTTON 150W

Why the interval matters as much as the numbers. The same three weeks give 98 seats a week averaged, or 122 a week if you take the last step alone. Neither is wrong; they answer different questions, and only one of them is a claim about the fortnight.

What a rising rate of change means. The steps are 25 and 122, so the rate is not constant — the curve is bending upward. A quantity whose rate of change is itself increasing cannot be described by any single average without saying which interval it is the average of.

Why the shape of a variety advance is known. Dowd has twenty-two years of it: three weeks flat, then vertical in the last fortnight. That is a fact about how people buy tickets rather than about this show. It is why a week-two decision is a decision made on the flat part.

TAKES AS READ

the three weekly figures are complete weeks · a function as a rule, its domain and what the domain rules out — taken as read

VERDICT 97W

The change is 147 seats over two weekly steps, so the average rate is 73.5 a week. The last step alone is 122. Both are honest and they answer different questions, which is the whole point: a rate of change is a property of an interval, not of a curve. What the two together tell you is that the rate is itself rising, so no single number describes the fortnight. Dividing by three rather than two is the common slip — three figures have two gaps between them — and it understates the rate by a third.

TAKEAWAY 16W

The same three numbers give two honest rates, and the interval is which one you mean.

1.2 Two lines that have to meet

THE HOUSE · A PERSON · CHOICE

SCENE 29W

Kettleby has the 1911 seat map and a proscenium narrowed by two feet three inches in 1958. Nobody has drawn a sightline since. Nine hundred seats are on sale.

GUIDE 65W

A sightline is a straight line from a seat to a point on stage. So a seat is not good or bad on its own. It is a seat that can or cannot see a particular place. Work out what has to be known before any seat can be called into question, and which of the four things on offer is the one nobody has.

ASK 13W

What has to be known before any seat can be called into question?

BEHIND THE BUTTON 151W

Why a seat is not the unit. The same seat sees a performer at centre stage and not one standing three metres left of centre. So the question is a pair: a seat, and a place somebody stands. That is what makes it two lines rather than a rating.

What the 1958 alteration did. Narrowing the proscenium by two feet three inches moved both edges of the frame inward. Every seat at the outer end of a row lost some of the stage, and the further forward the seat, the more it lost. That is why the front rows are the ones in question rather than the back.

Why the domain matters here. The function that matters takes a seat and a stage position and returns whether the line is clear. Its domain is every pair, and the pairs nobody has evaluated are the ones the show is about to sell.

TAKES AS READ

the 1958 alteration was symmetric about the centre line

VERDICT 97W

A sightline takes two things: a seat, and a place on stage. Without the second, no seat can be assessed at all, and the blocking is exactly what nobody has written down yet. That is why the question goes to the director before it goes to the seat map. The proscenium widths are known and are not sufficient — they say the frame moved without saying who lost what. The seat numbering is a naming scheme rather than a geometry, and the price a seat sold for is what the problem costs rather than whether it exists.

TAKEAWAY 20W

A sightline is a pair — a seat and a place somebody stands — not a property of a seat.

1.3 An arc, a radius and an angle

SCENE SHOP · A A PLACE · BALLPARK

SCENE 45W

Petra Novak, who runs the scene shop, is setting out a curved rostrum on the stage floor. The design gives a radius of 4.2 metres and an arc that turns through 1.15 radians. She needs the length of the front edge before she orders timber.

GUIDE 57W

An angle in radians is defined as the arc divided by the radius, so the arc is just the radius multiplied by the angle. That is the whole formula, and it is why radians have no conversion factor in them. Work out the front edge, and note what the same angle would give at a different radius.

ASK 8W

Estimate the length of the rostrum's front edge.

BEHIND THE BUTTON 148W

Why radians make this simple. Measured in degrees the same calculation needs a factor of pi over 180 every time. A radian is the angle whose arc equals its radius, so arc length is a multiplication and nothing else. Every formula in the course that mixes angles with lengths is written this way for that reason.

What the number is for. The front edge is faced with a single strip of ply bent round the curve, and it is ordered by length. Two hundred millimetres short is a joint in the middle of the front edge, on the piece the audience looks at all evening.

Why the radius has to come off the floor rather than the drawing. Novak sets out on the stage because the 1958 proscenium moved the centre line. A curve set out from the drawing's centre lands two feet three inches off the room's.

TAKES AS READ

the arc is a single circular curve of constant radius

VERDICT 103W

Radius times angle in radians is the arc: 4.2 times 1.15 is 4.83 metres. The degree figure is on the tile row because it is the same angle, and it cannot be used this way. Multiplying by 65.9 gives 277 metres, which is absurd enough to catch. It is exactly the mistake a calculator left in the wrong mode makes silently. The reason radians work here is definitional: a radian is the angle whose arc equals its radius, so the conversion factor is one. The diameter is the other trap, and it would give twice the answer, which is not absurd at all.

TAKEAWAY 16W

An angle in radians times a radius is an arc length, with nothing else in it.

END OF THE DAY 12W

A rate of change belongs to the interval it was measured over.

DAY 2 OF 15

An even step, and a doubling one

PLAN CARD 94W

Tuesday, and two lists have to be written before the first flying rehearsal. Simone Haigh, the stage manager, wants rehearsal calls on an even interval so forty-one people can see where the day ends. Kwame Osei, the head flyman, needs the trim heights for eleven battens, which are not evenly spaced because the grid is not. Marguerite Sallow wants both on one sheet by six. Today you build each sequence, and say which kind each one is. A list that looks even and is not is a list somebody works to and gets wrong.

BRIEFING 17W

Haigh needs a call sheet and Osei needs trim heights, and they are different kinds of sequence.

WHAT YOU DECIDE 15W

Build the rehearsal sequence and the batten trim heights, and tell the two kinds apart.

2.1 The same step, eleven times

FLY FLOOR · A A PLACE · VERIFY

SCENE 33W

Osei's trim heights start at 2.40 metres for the first batten and rise 0.34 metres for each one upstage. There are eleven battens. He needs the last one before he sets the counterweights.

GUIDE 59W

A sequence that adds the same amount each time is fixed by two numbers: where it starts and what it adds. The trap is the count. Going from the first term to the eleventh is ten steps, not eleven. One step of 0.34 metres is the difference between a piece that clears the grid and one that does not.

ASK 13W

Predict the eleventh trim height, then put a tape on it and report.

BEHIND THE BUTTON 137W

Why the trim heights rise at all. The grid is not level with the stage. Each batten upstage sits a little higher, so a cloth hung on the eleventh has to be flown further to clear the one in front of it. The rise is set by the ironwork and is genuinely constant.

Why the off-by-one is the whole risk. Eleven terms have ten gaps. Using eleven gives 6.14 metres instead of 5.80. That is a batten trimmed 340 millimetres high, enough that the cloth shows its bottom edge to the front rows.

What makes it arithmetic rather than geometric. The heights add a constant, they do not multiply by one. A geometric rise would put the eleventh batten at nearly forty metres, which is above the roof, and noticing that is faster than checking the arithmetic.

TAKES AS READ

the rise between battens is constant across all eleven · average rate of change over an interval — taken as read

VERDICT 87W

Eleven battens have ten gaps, so the prediction is 2.40 plus ten steps of 0.34: 5.80 metres. The tape says 5.74. Six centimetres over ten spacings is the ironwork rather than the arithmetic, and it is exactly the sort of drift a 1911 building accumulates. What the measurement rules out is the off-by-one. Eleven steps would have given 6.14, which is 400 millimetres above what the tape found. Of those millimetres, 340 would have been a cloth showing its bottom edge to the front rows all evening.

TAKEAWAY 12W

A sequence with n terms has $n - 1$ steps in it.

2.2 Sixteen pieces, each a little shorter

SCENE SHOP · A A PLACE · CHOICE

SCENE 39W

Novak has sixteen treads for a raked rostrum. The first is 3.60 metres and each one is 4 per cent shorter than the one below it. Quirke, the master carpenter, has written the order out as a constant subtraction.

GUIDE 48W

Read what the design says rather than what the order sheet says. Four per cent shorter than the one below is a multiplication, repeated; a constant subtraction is an addition, repeated. Work out where the two lists disagree, and by how much at the top of the rake.

ASK 11W

What is wrong with writing the treads as a constant subtraction?

BEHIND THE BUTTON 132W

Why the two look identical at first. Over the first two or three terms, four per cent of 3.60 is about 0.14 metres, so a constant subtraction of 0.14 fits almost exactly. The lists diverge slowly and then obviously, which is why the mistake survives the first few pieces being cut correctly.

Where they end up. Sixteen terms at four per cent leaves the last tread at 1.98 metres. Sixteen terms subtracting 0.14 leaves it at 1.50. Half a metre of tread, on a piece a performer stands on.

Which one the design means. A raked rostrum narrows in proportion because it is following a perspective, and a perspective is a ratio. That is a fact about the drawing rather than about arithmetic, and it is the reason to ask before ordering.

TAKES AS READ

the design's four per cent is per tread rather than overall · average rate of change over an interval — taken as read

VERDICT 98W

Four per cent of 3.60 is 0.144. So a constant subtraction of about 0.14 fits the first two or three treads almost exactly, and then drifts. The ratio takes four per cent of a steadily smaller number, while the subtraction keeps taking the same amount. By the sixteenth tread the two lists are 1.98 metres and 1.50 metres, which is half a metre of timber on a piece somebody stands on. The subtraction does not go negative in sixteen terms, so nothing about the order sheet looks absurd, and that is exactly why it would have been cut.

TAKEAWAY 17W

A constant proportional change is a ratio, and it only looks like a difference at the start.

2.3 Four stages, each times something

ORCHESTRA PIT AND SOUND · A PERSON · BALLPARK

SCENE 32W

Ilse Brand, the sound designer, has a signal through four stages, each multiplying it by 3.2: the microphone preamp, the desk channel, the group and the amplifier. The input is 0.004 volts.

GUIDE 49W

Gain stages multiply. Four of them at 3.2 each is not 12.8 times the signal, and the difference between adding and multiplying is the entire reason a chain like this clips. Work out what arrives at the amplifier, then compare it with what adding the gains would have suggested.

ASK 7W

Estimate the signal arriving at the amplifier.

BEHIND THE BUTTON 160W

Why every stage multiplies. Each one takes whatever arrives and scales it, so the output is the input times the product of the gains. That is a geometric sequence with the input as its first term. It is why gain is quoted in decibels: logarithms turn the multiplication into an addition a person can do in their head.

What clipping is. Every stage has a ceiling, and a chain that multiplies reaches it suddenly. Four stages of 3.2 is a factor of 105. So an input a little hotter than expected arrives at the amplifier far past what it can pass, and the result is distortion rather than loudness.

Why the fourth stage is where it shows. The first three multiply a small number and the fourth multiplies a large one. The same 3.2 is worth 0.009 volts at the first stage and 0.29 at the last. That is why gain is set from the front of the chain backwards.

TAKES AS READ

each stage passes its input to the next without limiting · arithmetic sequences: a constant difference — taken as read

VERDICT 98W

Four stages of 3.2 multiply to 104.9, so 4 millivolts becomes 420. Adding the gains gives 12.8 and an answer of 51 millivolts. That is a factor of eight out, in the direction that makes a chain look safe when it is not. That is what makes this a geometric sequence rather than an arithmetic one: each stage scales what arrives rather than adding to it. It also explains where the trouble appears. The same 3.2 is worth nine millivolts at the first stage and 290 at the last, so gain is always set from the front backwards.

TAKEAWAY 12W

Gains multiply, so a chain reaches its ceiling suddenly rather than gradually.

END OF THE DAY 13W

A constant difference and a constant ratio are two different kinds of list.

What the lantern is pointing at

PLAN CARD 111W

Wednesday. The rig has to be plotted before the bars go out on Thursday. Tunde Adeyemi, the lighting designer, works in bearings and spreads. He converts to positions afterwards. Nuno Ferreira, the chief electrician, has to hang the lamps on a bar. The bar positions on his plan are from 1911. Sallow wants the first look on Friday. Today you turn two measurements into an angle. Then you turn that angle into a spot on a bar. You also say which shape the advance sales are following. Any lamp hung on a wrong angle gets hung again. That happens with the whole company standing about. The notes session is at six.

BRIEFING 16W

Adeyemi's plot is in bearings and the bars are hung on a plan drawn in 1911.

WHAT YOU DECIDE 13W

Get an angle out of two measurements, and a beam onto a mark.

3.1 Six metres up, nine metres out

LIGHTING BOARD · A A PLACE · BALLPARK

SCENE 28W

The front-of-house bar is 6.2 metres above the stage floor and 9.4 metres out from the performer's mark. Adeyemi wants the down angle before Ferreira sets the yokes.

GUIDE 65W

Two sides of a right triangle fix the angle. The height above the mark is opposite the angle at the lantern, and the horizontal distance is adjacent to it. So the ratio of those two is the tangent, and the inverse tangent of it is the angle. Work it out, then say what the same lantern would need if the bar were two metres lower.

ASK 10W

Estimate the down angle from the bar to the mark.

BEHIND THE BUTTON 151W

Why this angle rather than another. A performer's face is lit by what arrives from above and in front, and the usual figure is somewhere near thirty-five to forty-five degrees down. Steeper puts the eyes in shadow; shallower puts the light in the audience's eyes and flattens the face.

Why the inverse function is needed. The ratio is the easy part; the angle has to be recovered from it, and that is what an inverse trigonometric function is for. It returns one angle from a restricted range, which is all that is wanted here because a lantern's down angle is between zero and ninety degrees by construction.

Why the numbers come off the room. The bar height is measured with a tape from the stage floor, and the distance from the mark is stepped out on the deck. The 1911 plan puts the bar 400 millimetres further out than it is.

TAKES AS READ

the mark is on the centre line of the beam · the unit circle, and sine and cosine as coordinates — taken as read

VERDICT 114W

The height is opposite the angle and the distance is adjacent to it. So the tangent is 6.2 over 9.4, which is 0.66, and the inverse tangent of that is 33.4 degrees. The arithmetic settles a rigging argument: 33 degrees is shallow for a face light, so either the bar comes in or the mark moves upstage. Using the straight-line throw of 11.3 metres as one of the two sides gives 33 degrees for the wrong reason. That length is the hypotenuse, and it belongs in a sine or a cosine rather than a tangent. The 45 degrees is what a designer would like and is not a measurement of anything in this room.

TAKEAWAY 16W

A ratio of two sides gives the tangent, and the inverse tangent gives the angle back.

3.2 Flat, then vertical

BOX OFFICE · A PERSON · CHOICE

SCENE 30W

Dowd has four weeks now: 41, 66, 188, 402. Sallow wants a forecast for the fortnight. A straight line through the four points misses the last one by ninety seats.

GUIDE 63W

Fit each candidate shape in your head and look at what it leaves over. A straight line has a constant difference, an exponential has a constant ratio, and a quadratic has a constant second difference. Work out which of the three the four numbers actually have. Treat the residuals — what each model missed by — as the evidence rather than the eye.

ASK 10W

Which family do the four weekly figures actually belong to?

BEHIND THE BUTTON 131W

What the differences say. The steps are 25, 122 and 214, which is not constant, so it is not linear. The ratios are 1.6, 2.8 and 2.1, which is not constant either. The second differences are 97 and 92, which very nearly are — and that is a quadratic.

Why residuals rather than appearance. Any smooth curve through four points looks plausible drawn by hand. The residual is the observed value minus the predicted one, and a model with a pattern in its residuals is the wrong shape however close it looks.

Why the choice matters more than the fit. All three models agree closely inside the four weeks and disagree wildly outside them, which is where the forecast is. The fortnight ahead is exactly the region the data cannot arbitrate.

TAKES AS READ

the four weekly figures are complete and comparable · average rate of change over an interval — taken as read

VERDICT 102W

The first differences are 25, 122 and 214, which rules out a straight line. The ratios are 1.6, 2.8 and 2.1, which rules out an exponential — a constant ratio is exactly what an exponential has. The second differences are 97 and 92, near enough constant to identify a quadratic. Rising every week makes a curve increasing rather than linear, and one doubling is not a constant ratio. Four points can identify a family, and the honest caveat is a different one. All three models agree inside the data and disagree wildly outside it, which is precisely where the forecast is wanted.

TAKEAWAY 13W

Each family of function has its own fingerprint in a table of values.

3.3 Where the frame cuts in

THE HOUSE · A A PLACE · PROTOCOL

SCENE 34W

The proscenium edge is 5.1 metres from the centre line. Kettleby is standing on the mark the show puts a performer, 2.8 metres left of centre and 3.4 metres upstage of the proscenium line.

GUIDE 63W

The seat, the proscenium edge and the performer are three points. Whether the seat can see the performer is whether the edge falls inside or outside the line between the other two. Work out the angle the sightline makes with the centre line, and the angle to the edge, and compare them. The one that is larger is what is in the way.

ASK 10W

Match each pair of measurements to the angle it gives.

BEHIND THE BUTTON 155W

Why a performer upstage is worse than one downstage. The further behind the proscenium line somebody stands, the more of the stage the frame cuts off for a seat at the side. The line from that seat has to pass the frame at a shallower angle. Three and a half metres upstage is a normal position for a solo spot.

What the comparison actually is. Two angles measured from the same point: the angle to the performer and the angle to the frame edge. If the frame's angle is the smaller one, the frame is inside the sightline and the seat sees the performer; if it is larger, it is not.

Why this is arithmetic rather than judgement. A seat either has a clear line or it does not, and the answer follows from four measured distances. What is a judgement is what to do about it, and that is tomorrow's argument rather than today's.

TAKES AS READ

the proscenium edge is a straight vertical line at that distance from centre · the unit circle, and sine and cosine as coordinates — taken as read

VERDICT 108W

Every angle here comes from two perpendicular distances, and the discipline is keeping track of which point each one is measured from. The performer's position is an offset and a depth. The frame's angle is measured from the seat, using the frame's offset and the seat's distance back. The seat's own bearing uses its own offset and the same depth. The fourth pair is the one that is not an angle at all. Two distances measured across the stage give the gap between the frame and the performer. That gap is 2.3 metres, and it says nothing about any seat until it is put beside a distance back.

TAKEAWAY 14W

Whether a seat sees a mark is two angles from the same point, compared.

3.4 An arc, a radius and an angle — Review

SCENE SHOP · A A PLACE · CALLBACK · VERIFY

SCENE 43W

Eamon Quirke, the master carpenter, is bending a steel handrail round the front of the new orchestra rail. The setting-out on the shop floor gives a radius of 3.6 metres and an angle of 1.45 radians. The steel is ordered by the metre.

GUIDE 68W

Multiply the radius by the angle in radians and you have the arc, with no conversion factor anywhere in it. Lock that figure before anything is chalked. Then have the curve set out full size and run a tape round it, and read what the floor says rather than what the drawing says. The gap between the two is the thing worth knowing before the steel is cut.

ASK 15W

Predict the length of the handrail, then run a tape round the setting-out and report.

BEHIND THE BUTTON 156W

Why the formula has nothing else in it. A radian is defined as the angle whose arc equals its radius, so the arc is the radius scaled by however many radians the curve turns through. In degrees the same calculation carries a factor of pi over 180, and a calculator in the wrong mode gives 299 metres here without complaining.

Why a tape is worth a prediction. The chalk line is a mathematical curve on a flat floor. The handrail is a real section with a thickness, bent round the outside of that line. So it follows a slightly larger radius than the one on the drawing.

What the gap costs if it is ignored. Steel comes in a stock length and is cut once. A rail ordered short means a joint in the middle of the front of the orchestra rail. It is at waist height, where the front rows lean on it all evening.

TAKES AS READ

the curve is a single arc of constant radius

VERDICT 83W

Radius times angle in radians is the arc, so the setting-out gives 3.6 by 1.45, or 5.22 metres. The tape says 5.31. The gap is nine centimetres, and it is geometry rather than error. The chalk line is the centre of the curve, and the handrail is bent round its outside face, sixty millimetres further out. Sixty millimetres of extra radius over 1.45 radians is nine centimetres of extra steel. Order to the chalk line and the rail is short at the joint.

TAKEAWAY 19W

An arc computed from a drawing is the length of the line, not of the thing bent round it.

END OF THE DAY 19W

Two sides of a right triangle fix an angle, and the angle is what a rigger can work to.

Four numbers describe a wave

BEFORE THE DAY — FOLLOW

Follow Novak round the set-out

Petra Novak, who runs the scene shop, sets out the curved rostrum on the stage floor itself rather than on paper, and she does not stop walking to explain it. Stay with her: the arc she is stepping out is the piece the flying sequence has to clear, and its radius is the number three later calculations use.

PLAN CARD 109W

Thursday. The band plays into the room for the first time at two. Ilse Brand, the sound designer, has a trace off the sound desk. It is a picture of the sound the band made. The pit cannot read it. Thandi Mbatha, the musical director, has nine players in the pit. The music in front of them was written for twenty-two. Meanwhile Dowd's returns are coming in, and Sallow wants them forecast. Today you describe a wave in the four numbers that fix it. You also put a shape on the set of returns. A waveform argued about in adjectives is a waveform nobody can set a fader to.

BRIEFING 14W

Brand has a trace off the desk and Dowd has three weeks of returns.

WHAT YOU DECIDE 15W

Read a waveform for its four parameters, and fit a decay to the ticket returns.

4.1 Tall, often, centred, and starting where

ORCHESTRA PIT AND SOUND · A A PLACE · BALLPARK

SCENE 26W

The trace runs between 0.18 and 0.62 volts. It repeats every 4.5 milliseconds. Its first upward crossing of the middle comes 0.9 milliseconds after the trigger.

GUIDE 63W

Four numbers fix any sine wave and each one comes off the trace directly. The midline is halfway between the top and the bottom. The amplitude is half the distance between them. The period is the repeat, and the phase is where the cycle starts relative to the trigger. Work out the first two and say what the other two are read from.

ASK 6W

Estimate the amplitude of the trace.

BEHIND THE BUTTON 157W

Why amplitude is half and not all. The amplitude is measured from the midline to the peak, so the peak-to-peak figure an oscilloscope shows is twice it. This is the most common misreading of a trace and it doubles every level derived from it.

Why the midline is not zero here. The signal is riding on a direct offset of 0.40 volts, which is a fault rather than a feature: something upstream is passing a standing voltage. It does not change the sound and it does eat headroom, and it is visible only because the midline was read rather than assumed.

Why the phase matters to Mbatha rather than to Brand. Two instruments a fraction of a cycle apart do not sound bigger, they sound wrong, and the fix is timing. The phase shift is the number that says how far apart they are in the only units that matter, which is a fraction of a period.

TAKES AS READ

the trace is a single sinusoid rather than a sum of several · the unit circle, and sine and cosine as coordinates — taken as read

VERDICT 96W

The peak-to-peak span is 0.44 volts, so the amplitude — measured from the midline to the peak — is half of it: 0.22. Reading 0.44 as the amplitude is the standard error and doubles every level derived from the trace. The more interesting number is the one that falls out for free. The midline is 0.40 volts rather than zero, so the signal is riding on a standing offset. It does not change what anybody hears and it eats headroom, and it is only visible because the midline was read rather than assumed to be zero.

TAKEAWAY 13W

Amplitude is measured from the midline, not from the bottom of the trace.

4.2 The same fraction each week

BOX OFFICE · A PERSON · BALLPARK

SCENE 27W

Returned tickets: 84 in the first week after they went on sale, then 59, then 41. Dowd wants the fourth week before she releases the held seats.

GUIDE 51W

Look at the ratios rather than the differences. Each week's returns divided by the week before gives roughly the same number, which is what an exponential is. Work out the ratio, apply it once more, and say what the model does and does not entitle anybody to claim about week eight.

ASK 7W

Estimate next week's returns from the ratio.

BEHIND THE BUTTON 147W

Why returns decay proportionally. The people most likely to return a ticket do it soon after buying, so each week draws on a shrinking pool of recent buyers. That produces a constant fraction rather than a constant number, which is the signature of an exponential.

Why the difference test fails here. The drops are 25 and 18, which is not constant, so a straight line is out. The ratios are 0.70 and 0.69, which is about as constant as three real numbers get, and it is the ratio that identifies the family.

What it cannot say. The model is fitted over three weeks and the fortnight ahead is outside them. Returns will not follow this curve to zero. There is a floor of cancellations that happen for reasons unrelated to buying. And a model quoted past its data is a claim about a mechanism nobody has checked.

TAKES AS READ

the three weekly return figures cover complete weeks · geometric sequences: a constant ratio — taken as read

VERDICT 114W

The ratios are 0.70 and 0.69, so the sensible model multiplies by about 0.70 each week: 41 becomes 29. Subtracting the last drop of 18 gives 23 instead, and the two answers diverge fast. By week six the ratio model says 14, and the subtraction says nothing at all, because it has gone negative. That is the practical test between the families and it is worth doing on three numbers. What the model does not license is week eight. It is fitted on three weeks, and returns have a floor made of cancellations that happen for their own reasons. A curve quoted past its data is a claim about a mechanism nobody has checked.

TAKEAWAY 12W

A constant ratio identifies an exponential; a constant difference identifies a line.

4.3 **Twice as far, a quarter as bright**

LIGHTING BOARD · A A PLACE · CHOICE

SCENE 41W

Ferreira has to move a unit off the front bar because the flying sequence needs the space. The new position is twice as far from the mark. Adeyemi says the level goes to a quarter; Sallow has been told it halves.

GUIDE 66W

Ask what the function of distance actually is before arguing about the factor. Light from a point source spreads over a sphere. The area of a sphere goes as the square of the radius, so the level at a surface goes as one over the distance squared. Then say what that means for the two claims in the room, and what could be done about it.

ASK 12W

What does doubling the throw do to the level at the mark?

BEHIND THE BUTTON 154W

Why the square rather than the distance. The same light spreads over a surface that grows as the square of the radius, so it is diluted by that square. Doubling the distance quadruples the area and quarters the level. This is not specific to light: it is what happens to anything radiating outward from a point.

What can be done about it. A narrower beam angle puts the same output on a smaller patch, which recovers level at the cost of coverage. So a unit moved back can be re-lensed rather than replaced. What cannot be recovered is the softness that comes with a wide beam close in.

Why the mistake is so common. Intuition is linear. Twice as far sounding like half as bright is one of the most reliable wrong answers in a technical rehearsal. The correction is a factor of two on a number somebody has already set a fader to.

TAKES AS READ

the unit can be treated as a point source at these distances · a function as a rule, its domain and what the domain rules out — taken as read

VERDICT 121W

Light from a point spreads over a sphere, and a sphere's area goes as the square of its radius. So twice the distance is four times the area, and a quarter of the level. Halving is the intuitive answer and it is wrong by a factor of two on a number somebody has already set a fader to. Keeping the beam angle unchanged is exactly the case being described — the angle is what makes the patch bigger as the unit goes back. Colour costs level through the filter and does so at any distance, so it is a separate multiplication rather than an explanation of this one. What recovers the level is a narrower lens, at the price of coverage.

TAKEAWAY 13W

A quantity spreading over a sphere falls as the square of the distance.

4.4 The same step, eleven times — Review

FLY FLOOR · A A PLACE · CALLBACK · SEQUENCE

SCENE 41W

Simone Haigh, the stage manager, has four hemp sets to dead-hang before the run-through. The drawing gives each one as a starting height and a step rather than as a height. Kwame Osei wants them in order before anybody goes up.

GUIDE 71W

Work each set out before you order any of them. A term is the starting height plus the step added one time fewer than the term number. So the ninth batten of a set has eight steps in it. None of the three numbers settles the order on its own. The set with the lowest start does not finish lowest, and the set with the biggest step does not finish highest.

ASK 14W

Put the four hemp sets in order of the trim height each rule gives.

BEHIND THE BUTTON 154W

Why the drawing is written this way. A hemp set is a run of battens on one ironwork rake, so the rake is a start and a constant rise. Writing the rule rather than eleven heights is shorter and it survives the set being extended, which is why the fly floor has kept the convention since 1911.

Where the off-by-one lands. Nine battens have eight gaps between them. Using the term number as the step count adds one extra rise to every set. The steps here differ, so it adds a different amount to each. That is what changes the order rather than just the heights.

Why the order is the thing wanted. Osei is not hanging to a number, he is hanging four pieces that pass one another in the same 12 metres of grid. What decides whether they foul is which is under which, and that is settled before anybody weighs anything.

TAKES AS READ

the rise between battens is constant along each set

VERDICT 88W

A term is the start plus one step fewer than the term number. Three steps to the fourth, six to the seventh, eight to the ninth, eleven to the twelfth. That gives 3.76, 3.90, 4.04 and 4.25 metres, so Set 3 hangs lowest and Set 12 highest. None of the three numbers decides it alone. Set 19 starts lowest and finishes third, and Set 3 has the highest start and the lowest trim. Using the term number itself as the step count swaps Set 7 and Set 19.

TAKEAWAY 20W

A start, a step and a term number are three separate handles, and no two of them decide a height.

END OF THE DAY 9W

Amplitude, midline, period and phase fix a sinusoid completely.

Adding what multiplies

PLAN CARD 99W

Friday. The first technical run is at four. Brand has a chain of gains, and gains multiply. Her desk is marked in decibels. Mbatha wants two channels matched by ear, not by number. Brand will not do it that way. Adeyemi's rig plot is a list of bearings and distances. Ferreira has to turn that list into positions on a bar. Today you use logarithms to add what would otherwise multiply. You also convert a beam from a bearing to a coordinate. Two people describe the same rig in different coordinates. They argue for an hour and hang nothing.

BRIEFING 12W

Brand's desk is calibrated in decibels and Adeyemi's plot is in bearings.

WHAT YOU DECIDE 12W

Turn the gain chain into decibels and the beam plot into coordinates.

5.1 Why a fader is a logarithm

ORCHESTRA PIT AND SOUND · A A PLACE · BALANCE

SCENE 30W

The chain multiplies by 3.2, then 4.0, then 2.5, then 8.0. Brand wants each stage marked on the desk in decibels so the crew can add them in their heads.

GUIDE 54W

A product of four factors becomes a sum of four logarithms. Take the log of each gain, multiply by twenty to get voltage decibels, and add. Then check the answer against the log of the product. The whole point of the rule is that those two are the same number arrived at two ways.

ASK 12W

Read what you need and report the chain's total gain in decibels.

BEHIND THE BUTTON 150W

Why twenty rather than ten. For a voltage ratio the decibel is twenty times the log, because power goes as the square of voltage and the ten-times-log definition is written for power. Mixing the two is a factor of two on every number, which is four decibels on a typical stage gain.

What the rule buys in practice. Four multiplications with a calculator become four additions in the head. A crew can then talk about a stage as plus ten or minus six rather than as times three point two. Every fader, every trim and every pad on the desk is marked that way for this reason.

Why the check matters. The sum of the logs and the log of the product are equal by the rule, so computing both is a free verification. A discrepancy means a gain was written down wrong, which is what the check is for.

TAKES AS READ

the gains are voltage ratios rather than power ratios · logarithms as inverse exponentials — taken as read

VERDICT 88W

Each stage multiplies. A log turns that into a sum. So the four stages add to 48.2 dB rather than being multiplied out to 256. The term that hides is the pad in the mike barrel. Ten dB, switched in at the capsule, and nothing on the desk shows it. It is also why the channel needed so much gain. The total is 38 dB, not 48. The fader is the other trap. At unity it is a position rather than a gain, so counting it adds nothing.

TAKEAWAY 19W

The sum of the logs equals the log of the product, which is why gains add on a desk.

5.2 From the centre out

LIGHTING BOARD · A PERSON · BALLPARK

SCENE 35W

Adeyemi's plot puts a unit 7.4 metres from the centre of the proscenium at a bearing of 38 degrees off the centre line. Ferreira works in metres left or right of centre and metres upstage.

GUIDE 57W

A bearing and a distance are one description of a point and two perpendicular offsets are another. The conversion is a right triangle: the distance is the hypotenuse, the bearing is the angle at the centre, and the two offsets are the sides. Work out both offsets, and say which of them the rigger actually needs first.

ASK 8W

Estimate how far off centre the unit hangs.

BEHIND THE BUTTON 139W

Why a designer works in polar and a rigger in rectangular. A lantern is aimed, so its natural description is a direction and a throw. A bar is a straight line with positions along it, so its natural description is a distance from centre. Both are right and the conversion is the interface between two trades.

Which offset matters first. The across-stage figure is where on the bar the unit hangs, which has to be right before anything is bolted. The upstage figure decides which bar, and there are only four to choose from, so it is coarser and easier to fix.

What goes wrong without the conversion. Two people describing the same rig in different coordinates agree about everything and hang nothing. Or worse, they hang it 900 millimetres out because a bearing was read as an offset.

TAKES AS READ

the bearing is measured from the centre line toward stage left · polar coordinates, and a beam described as r of theta — taken as read

VERDICT 108W

The distance is the hypotenuse and the bearing is the angle at the centre, so the across-stage offset is 7.4 times the sine of 38 degrees: 4.56 metres. Using the cosine gives 5.83, which is the upstage offset. That is a perfectly real number for a different question, and the reason the two sit on the same tile row. Getting them the wrong way round hangs the unit 1.3 metres out of position. On a front bar that is the difference between lighting a performer and lighting the person beside them. The complement of the angle is on the row because using it swaps the two offsets silently.

TAKEAWAY 13W

A bearing and a distance convert to two offsets through one right triangle.

5.3 Not half each

FLY FLOOR · A A PLACE · BALLPARK

SCENE 37W

A cloth weighing 180 kilograms hangs from two lines. The grid is narrower than the stage, so each line leans 22 degrees off the vertical. Osei will not lift it until the per-line figure is written down.

GUIDE 56W

A line pulls along its own length, so only part of that pull is vertical. Split each line's tension into a vertical part, which carries the weight, and a horizontal part, which does not. Then work out what each line has to pull for the two vertical parts to add up to 180 kilograms of load.

ASK 9W

Estimate the tension in each of the two lines.

BEHIND THE BUTTON 147W

Why leaning costs. The vertical components have to sum to the weight whatever the angle. So as the lines lean further, each one has to pull harder to keep its vertical part the same. At 22 degrees the penalty is about eight per cent; at 60 degrees each line pulls the whole weight.

Where the horizontal parts go. They point toward each other and cancel at the piece, which is why it hangs still. They do not cancel at the grid: each line puts a sideways pull into the ironwork, which is what the 1958 conversion never accounted for.

Why Osei wants it written. A per-line tension is what a line's rating is compared against, and half the weight is the number people say. The difference between 90 and 97 kilograms is small; the difference at the angles this grid forces on a wide cloth is not.

TAKES AS READ

the two lines are symmetric about the piece's centre of mass · right-triangle ratios, and finding an angle from two sides — taken as read

VERDICT 107W

Each line's vertical component is its tension times the cosine of its lean, and the two vertical components have to add to 180 kilograms. So each line pulls 180 over twice the cosine of 22 degrees, which is 97.1 kilograms. Halving the weight gives 90 and assumes the lines are vertical, which they are not, because the grid is narrower than the stage. Eight per cent sounds ignorable, and the point is the shape of the relationship. At 45 degrees each line pulls 127 kilograms, and at 60 it pulls the whole 180. The horizontal components cancel at the cloth and do not cancel at the grid.

TAKEAWAY 11W

Only the vertical component of a leaning line carries the load.

5.4 Off the flat part — Review

BOX OFFICE · A A PLACE · CALLBACK · PROTOCOL

SCENE 45W

Ruth Dowd, the box office manager, has the telephone log from the on-sale week. It reads 120 bookings by the end of day one, 168 by day three, 300 by day five, 540 by day seven. Four people have quoted four different rates off it.

GUIDE 64W

Take each interval on its own. Subtract the earlier reading from the later one, divide by the number of days between the two, and pair the interval with the figure it gives. All four are honest arithmetic about one log. The count to watch is days rather than readings, because four readings across the week have six days between the first and the last.

ASK 12W

Match each interval of the booking log to the rate it gives.

BEHIND THE BUTTON 164W

Why one table gives four rates. A rate of change is a property of an interval rather than of a curve, so every pair of readings in the log has its own. None of them is more correct than another. What differs is the stretch of the week each one describes, and that is the part a quoted figure usually leaves out.

What the spread across these four says. Twenty-four a day at the start and 120 a day at the end is a rate that has multiplied by five inside one week. A quantity whose rate of change is itself rising cannot be summarised by any single average. That is why Dowd keeps the log by day rather than by week.

Why the staffing argument turns on this. Two clerks were rostered against the week's average of 70 a day. The last pair of days ran at 120 and the queue reached the street, and the fortnight ahead is steeper than either figure.

TAKES AS READ

the log is cumulative and each reading is taken at the end of that day

VERDICT 85W

Each interval is a difference divided by the days between the two readings, so one log gives four honest rates. The whole week averages 70 a day. The opening pair gives 24, the last pair 120, and the middle pair 66. Nothing is inconsistent: the rate is rising, and each figure is the average across the stretch it was taken from. The common slip is counting readings rather than days — four readings across the week, but six days between the first and the last.

TAKEAWAY 21W

A rate belongs to the interval it was measured over, so one table gives as many rates as it has intervals.

END OF THE DAY 11W

Logarithms turn multiplication into addition, which is what a fader is.

How often, how loud, how hard

PLAN CARD 98W

Saturday. Douglas Pell, the licensing officer, has asked for sound levels measured at the seat. Not at the desk, where they are easy to read. Haigh needs the period of the revolve. Then the cue can be called on a count instead of on a look. Osei has a second cloth to fly. He has the first cloth's figure to compare it against. Today you work out three numbers the licence file needs. A level at the desk and a level at a seat are two different things. Only one of them is what an audience actually gets.

BRIEFING 13W

Pell wants sound levels at the seat and Haigh wants the revolve timed.

WHAT YOU DECIDE 17W

Set a level against the licence, a period against a cue, and a resultant against a rating.

6.1 Ten times the log of it

ORCHESTRA PIT AND SOUND · A A PLACE · BALLPARK

SCENE 37W

The meter at row F reads an intensity of 3.2×10^{-4} watts per square metre. The reference for the scale is 1.0×10^{-12} watts per square metre. The licence caps the house at 103 decibels.

GUIDE 51W

A decibel level is ten times the logarithm of the measured intensity divided by the reference. Work out the ratio first, then the logarithm, then the ten. Compare what comes out with the licence cap, and note how much the whole argument turns on the scale being logarithmic rather than linear.

ASK 8W

Estimate the level at row F in decibels.

BEHIND THE BUTTON 91W

Why the reference is in the formula at all. A decibel is a ratio, so it needs something to be a ratio of. The standard reference is the quietest audible intensity, which is why the numbers come out between zero and about 130 for everything a person meets.

What the cap means in intensity. Three decibels is a doubling, so a house at 106 is twice the intensity of one at 103, not three per cent more. That is what makes an argument about a few decibels an argument about factors.

TAKES AS READ

the meter reading is an average over the passage rather than a peak · log properties, and why a product becomes a sum — taken as read

VERDICT 98W

The ratio is 3.2×10^{8}. Its logarithm is 8.51, and ten times that is 85 decibels. The margin to the cap is the part worth saying properly. Eighteen decibels is not eighteen per cent. It is a factor of about sixty in intensity, so the house has a great deal of headroom rather than a little. Using twenty as the multiplier gives 170. That is the voltage definition applied to an intensity, and it is above the threshold of pain — absurd enough to catch. Knowing which definition belongs to which quantity is what makes it catchable.

TAKEAWAY 14W

Ten decibels is a factor of ten, and three is a factor of two.

6.2 Two pi over the coefficient

LIGHTING BOARD · A A PLACE · VERIFY

SCENE 35W

The chase on the board runs as a sinusoid in the intensity of each circuit, written with an inside coefficient of 0.42 per second. Haigh needs the period to call the cue on a count.

GUIDE 48W

The coefficient inside the function is what squeezes or stretches the cycle, and the period is two pi divided by it. Work it out in seconds, then say what a larger coefficient would do — because the relationship is inverse and the intuition usually runs the wrong way.

ASK 13W

Predict the period from the coefficient, then run the chase and time it.

BEHIND THE BUTTON 89W

Why the relationship is inverse. The function completes a cycle when its argument advances by two pi. The argument advances B times as fast as the input, so a larger B means a shorter cycle. Doubling the coefficient halves the period.

What a count is for. A cue called on a count rather than on a look is repeatable. A period in seconds converts to a count at whatever tempo the band is playing. That conversion is why the number has to be a period and not a description.

TAKES AS READ

the chase runs at a constant coefficient through the cue · a sinusoid: amplitude, midline, period and phase — taken as read

VERDICT 92W

A cycle finishes when the argument advances by two pi. It advances 0.42 times as fast as the clock, so the period is 6.283 over 0.42: fifteen seconds. The board runs it at 15.2, which is the software doing what it was told. The relationship is inverse, which is where intuition fails — doubling the coefficient gives seven and a half seconds rather than thirty. Using pi instead of two pi halves the answer, and would have put the cue in the wrong bar. The measurement is what would have caught it.

TAKEAWAY 14W

A larger coefficient inside the function makes a shorter cycle, not a longer one.

6.3 What cancels at the cloth and not at the grid

FLY FLOOR · A PERSON · CHOICE

SCENE 31W

Yesterday's cloth hangs on two lines at 22 degrees, each pulling 97 kilograms. Before he adds a second cloth, Osei wants the sideways force each line puts into the grid ironwork.

GUIDE 54W

Split each tension into its two components again, but keep the horizontal one this time. At the cloth the two horizontal parts point at each other and cancel. At the grid they do not cancel. They act at two different points, and that is the load the 1958 conversion never had a figure for.

ASK 9W

What does the sideways component of each line do?

BEHIND THE BUTTON 88W

Why cancellation depends on where you stand. Two equal and opposite forces at the same point sum to nothing. The same two forces applied at points four metres apart put the structure between them into tension. Nothing about the piece hanging still says otherwise.

What the ironwork was designed for. Vertical load, which is what a grid over a stage-width flying system carries. The Ellery's grid is narrower than its stage. So every wide piece leans its lines, and adds a sideways component the drawings do not mention.

TAKES AS READ

both lines lean by the same angle · vectors: components, magnitude and direction — taken as read

VERDICT 106W

Each line pulls 97 kilograms along its own length. At 22 degrees the sideways part of that is about 36 kilograms. The two sideways parts point toward each other. So at the cloth they cancel and it hangs still, which is exactly why the problem is invisible. At the grid they are applied four metres apart. Two opposed forces at separated points put the structure between them in tension. Nothing about the piece being motionless says anything about what the ironwork carries. It does not add to the vertical load, which is a perpendicular direction. Unequal angles would make it worse rather than make it exist.

TAKEAWAY 15W

Equal and opposite forces cancel at a point and load the structure between two points.

END OF THE DAY 17W

A level is a ratio on a logarithmic scale, and a period is what one cycle costs.

DAY 7 OF 15

The room is the instrument

PLAN CARD 103W

Sunday. The seats are on a lorry that arrives Tuesday. Brand can hear what the empty house does to a sound. She cannot put a number on it until she measures the decay. Mbatha wants the pit rail lowered. That is a building job, and Novak would have to start it today. Pell wants the reverberation figure in the file whichever way it comes out. Today you measure a decay in the empty house. Then you say what it will be with eight hundred and sixty people sitting in it. A room measured empty is a measurement of a building nobody performs in.

BRIEFING 15W

The house rings, the seats arrive on Tuesday, and Pell wants a figure either way.

WHAT YOU DECIDE 16W

Measure what the empty house does to sound, and what the seats will do about it.

7.1 The same fraction each half second

THE HOUSE · A A PLACE · VERIFY

SCENE 29W

Brand claps once in the empty house. The level is 92 decibels at the clap and falls by 28 per cent of its remaining sound energy every half second.

GUIDE 48W

A decay that removes the same fraction each interval is a repeated multiplication. Work out what fraction of the original energy is left after eight half-seconds. Then say what that means for the reverberation time, which is the interval over which a room loses almost all of it.

ASK 12W

Predict what is left after four seconds, then clap and measure it.

BEHIND THE BUTTON 92W

Why a proportional decay never reaches zero. Each step takes a fraction of what remains, so there is always something left. In practice the sound falls below what anybody can hear. So reverberation time is defined against a stated drop rather than against silence.

Why the seats change it. Sound energy leaves a room by being absorbed, and an empty seat absorbs far less than an occupied one. Eight hundred and sixty people are the largest absorber the Ellery will ever contain, and their absence is why the room is currently unusable.

TAKES AS READ

the decay fraction is constant across the interval measured · geometric sequences: a constant ratio — taken as read

VERDICT 112W

Each half second leaves 72 per cent of what was there, and eight of those is 0.72 to the eighth: 7.2 per cent. The measurement gives 8.1. That is close enough to confirm the model, and far enough above nothing to be plainly audible from row F. It is what an empty variety house sounds like. Using the lost fraction instead gives 0.28 to the eighth, a ten-millionth, which would describe a recording studio. Multiplying 0.28 by eight is the linear reflex, and it gives more than two: more energy than the clap had. That is the fastest way to notice the model is a repeated multiplication rather than a repeated subtraction.

TAKEAWAY 13W

A proportional loss leaves a shrinking amount each step and never quite finishes.

7.2 Three inches, and what it costs

SCENE SHOP · A A PLACE · CHOICE

SCENE 36W

Mbatha wants the pit rail three inches lower. Novak has one day of shop time left before the seats arrive, and the rail is also a sightline for row A and a reflector for the band.

GUIDE 49W

One change, three consequences, and they are measured in three different units. Work out what each of them is actually a function of, and which of the three can be answered from numbers already in the file. The answer is which measurement to take rather than what to do.

ASK 9W

What should be measured before the rail is cut?

BEHIND THE BUTTON 89W

Why three inches is not a small change here. Row A sits 1.1 metres behind the rail, so three inches of rail height is nearly two degrees of sightline for that row. The rail is the top edge of what those seats see over.

What the rail does for the band. It reflects sound back at the players, which is how nine people covering an arrangement for twenty-two hear each other. Lowering it improves the house and removes some of that, which is Mbatha asking for two contradictory things.

TAKES AS READ

the shop has one working day before the seats arrive · average rate of change over an interval — taken as read

VERDICT 109W

Row A is the consequence that cannot be undone. Sound is adjustable — a fold-back for the band, or a reflector added later — and shop time is a scheduling fact rather than a measurement. But a seat that cannot see over a rail is a refund. The rail is the top edge of what row A looks over, at nearly two degrees per three inches from 1.1 metres back. Measuring the band's drop is worth doing and it is a level that can be recovered by other means. What the rail weighs is a construction detail with no bearing on any of the three questions in the room.

TAKEAWAY 11W

One change with three consequences needs three measurements, not one opinion.

7.3 Where it stops being one

BOX OFFICE · A PERSON · CHOICE

SCENE 26W

The quadratic through four weeks of advance sales fits closely. Extended to week eight it predicts 2,140 seats sold, in a house that holds nine hundred.

GUIDE 52W

Take the model at its word and see where the word breaks. Work out what the prediction says and compare it with something the model cannot know, then say which of the four statements about it is the honest one. A model that fits its data is a claim about that data.

ASK 11W

What is the honest thing to say about the week-eight figure?

BEHIND THE BUTTON 84W

What extrapolation actually assumes. That the mechanism producing the numbers keeps working the same way outside the range measured. Here it cannot: the house has nine hundred seats, so sales must flatten no matter what the curve does.

Why the fit is not the problem. The quadratic is the right family inside the four weeks, and its residuals are small. Choosing the right shape and then using it outside the data are two separate steps, and the second one is where this goes wrong.

TAKES AS READ

the house capacity is fixed at nine hundred · choosing a model, and what a residual says about the choice — taken as read

VERDICT 110W

Two things are separately true. Inside the four weeks the quadratic is the right family and its residuals are small, so the fit is not in question. Outside them the model assumes a mechanism that keeps working, and it cannot here, because there are nine hundred seats. So the honest sentence keeps the fit and refuses the forecast. Saying the family was wrong throws away a good description of what happened. Saying sales will reach nine hundred and stop is a different model nobody has fitted — plausible, and not what these four numbers say. And more data does not fix an extrapolation; it only moves where the extrapolation starts.

TAKEAWAY 11W

A model fitted inside a range says nothing certain outside it.

7.4 Four stages, each times something — Review

ORCHESTRA PIT AND SOUND · A A PLACE · CALLBACK · PROTOCOL

SCENE 38W

The pit foldback runs through three stages of 2.5 from a 0.006 volt pickup, and Thandi Mbatha, the musical director, cannot hear the cello. Ilse Brand has four changes on the desk and wants them thought through first.

GUIDE 51W

A chain that multiplies has three handles and they do different things: what goes in, the factor each stage applies, and how many stages there are. Take each proposed change, ask which of the three it touches, and pair it with the effect. One of the four touches none of them.

ASK 15W

Match each proposed change to what it does to the level at the foldback amplifier.

BEHIND THE BUTTON 158W

Why the three handles are separate. The output is the input times the factor raised to the number of stages. Changing the input scales everything by the same amount. Changing the number of stages adds or removes a whole factor. Changing the factor itself is the one that compounds, because it is raised to a power.

Why halving is worth more than it looks. Halving one gain halves the output. Halving all three divides it by eight, because each half is a factor and the factors multiply. That is the same arithmetic that makes a chain clip suddenly, run the other way.

What reordering does change. The level at the amplifier is the same whatever order the stages come in, because a product does not depend on its order. What moves is the noise. A stage that adds hiss early has that hiss amplified by everything after it, which is why the quietest gain sits at the front.

TAKES AS READ

each stage passes its input to the next without limiting

VERDICT 82W

A chain of gain stages is a geometric sequence, so it has three separate handles. Adding a stage adds a term, and the output picks up one more factor of 2.5. Halving all three gains divides the product by two cubed, which is an eighth. Doubling the pickup changes the first term and leaves the ratio alone, so everything downstream doubles with it. Reordering the stages changes nothing at all, because a product does not depend on the order of its factors.

TAKEAWAY 19W

What goes in, the factor and the number of stages are three handles, and only one of them compounds.

END OF THE DAY 13W

A quantity falling by the same fraction each interval never quite reaches zero.

What arrives at the performer

BEFORE THE DAY — HUNT

Six lanterns went out with the 1974 sale

The rig plot lists six units that are not on the bars, and Nuno Ferreira, the chief electrician, thinks they are still in the building rather than sold. Find all six. Every one you turn up is a unit the budget does not have to hire, and the plot has to be right before the board is plotted.

PLAN CARD 106W

Monday. The first full look is at seven. Adeyemi's plot is drawn. Ferreira has not yet put a meter under any of it. That is where every rig argument starts. Haigh has a cue that has to fire when the revolve reaches a mark. The chase reaches that level twice in every cycle. Today you work out what light arrives at the performer. You also find both times the cue could go, and which seats the walked survey condemns. A trigonometric equation with one answer written down has usually had a second one thrown away. A cue called on the wrong crossing fires fifteen seconds late.

BRIEFING 15W

The first look is tonight and the plot has not been checked against a meter.

WHAT YOU DECIDE 16W

Get a level onto a face, and find both angles a cue could be called at.

8.1 Four pi d squared

LIGHTING BOARD · A A PLACE · BALLPARK

SCENE 27W

The unit puts out 9,600 lumens. The mark is 11.3 metres away along the beam. Ferreira wants the figure before he puts a meter under it tonight.

GUIDE 55W

Light leaving a point spreads over a sphere, and the area of that sphere is four pi times the radius squared. Divide the output by that area to get what arrives per square metre. Then compare it with what the same unit gives at half the distance, because the factor is the part worth carrying.

ASK 7W

Estimate the illuminance arriving at the mark.

BEHIND THE BUTTON 107W

Why this is an estimate rather than a measurement. A real lantern is not a point and does not radiate in all directions. A lens puts most of the output into a cone, so the actual level is several times this figure. The spherical calculation gives the floor, which is why Ferreira meters it rather than trusting it.

What the square does to a rigging decision. Half the distance is four times the level, so a unit moved from eleven metres to five and a half gains two stops of light. That is why front-of-house positions are argued about more than any other part of a rig.

TAKES AS READ

the unit can be treated as radiating into a sphere for a floor estimate · a function as a rule, its domain and what the domain rules out — taken as read

VERDICT 116W

The sphere at 11.3 metres has an area of 1,604 square metres, so 9,600 lumens spread over it is 6 lux. That is a floor rather than a prediction. A lantern's lens puts most of its output into a narrow cone instead of a sphere, which is exactly why Ferreira meters at the mark. What the calculation is genuinely for is the exponent. Halving the distance quarters the area and quadruples the level. So moving a unit from eleven metres to five and a half is worth two stops, and that is why front-of-house positions are fought over. Dropping the four pi gives 75 lux and an answer that flatters the rig by more than tenfold.

TAKEAWAY 18W

Output over four pi d squared is the floor, and a lens is what puts it above that.

8.2 Twice a cycle

ORCHESTRA PIT AND SOUND · A PERSON · CHOICE

SCENE 34W

The chase rises and falls on a fifteen-second cycle between 10 and 90 per cent. Haigh's cue has to fire when it passes 70 per cent. The board's inverse function gives her one time.

GUIDE 52W

A sinusoid passes any level between its extremes twice in every cycle — once going up and once coming down. The inverse function returns one of the two, because it is restricted to a single range. Work out where the other one is and say which of them the cue should use.

ASK 11W

Where is the second time the chase passes 70 per cent?

BEHIND THE BUTTON 91W

Why an inverse trigonometric function returns one answer. Sine is not one-to-one, so it has no inverse until its domain is restricted, and the restriction is what makes the inverse a function. The cost of that convenience is that the other solutions have to be recovered by hand.

How to recover the second one. Within one cycle, the two times a rising-then-falling wave crosses a level are symmetric about the peak. So the second is as far after the peak as the first is before it, and both repeat every period.

TAKES AS READ

the chase runs one full cycle per cue · inverse trigonometric functions, and the range they are restricted to — taken as read · period from the coefficient inside the function — taken as read

VERDICT 118W

A rising-then-falling wave crosses any level between its extremes twice, and the two crossings are symmetric about the peak. So the second is as far after the peak as the first is before it. Half a period after the first is the general shape of the mistake. That spacing is only right for a crossing of the midline, and 70 per cent is well above it. One full period later is the same crossing in the next cycle rather than the second crossing in this one. For a cue that means firing fifteen seconds late. And there is certainly a second crossing, because the wave has to come back down; the inverse function simply does not return it.

TAKEAWAY 16W

A sinusoid crosses any interior level twice a cycle, and the inverse gives one of them.

8.3 Six rows, two seats each

THE HOUSE · A A PLACE · CHOICE

SCENE 34W

Kettleby has walked the front six rows with a marker on the performer's position. Rows A to F each lose the two outermost seats on the stage-left side. Sallow has nine hundred on sale.

GUIDE 63W

A sightline is two lines that either meet on the stage or do not, which is a pair of equations with a solution or without one. Work out what the walked measurement establishes. Then say what the honest description of a condemned seat is. The difference between a bad seat and a seat that cannot see decides whether it is sold at all.

ASK 6W

What does the walked survey establish?

BEHIND THE BUTTON 98W

What no solution looks like here. The line from the seat past the proscenium edge and the line to the performer do not intersect anywhere on stage. There is no point the seat can see the performer from. That is a geometric fact rather than a matter of degree.

Why twelve seats and not six or twenty-four. Both sides of the house are symmetric about the centre line, so the count is doubled. But the marker was on a stage-left position, and a stage-right mark would condemn the mirror pair instead. Twelve is the count for this blocking.

TAKES AS READ

the marker was placed where the show puts the performer · matrices as transformations of the plane — taken as read

VERDICT 100W

The survey establishes a specific thing: for this blocking, twelve seats have no line to the performer's mark. That is a pair of lines with no intersection on stage, rather than a bad view. The qualification matters: move the mark and the twelve seats change. Calling them poor-view seats invites selling them at a discount for a problem that is total rather than partial. Condemning the whole front six rows is a much larger claim than twelve seats and would cost about ninety. Widening the proscenium is undoing 1958 in six days, which is not a finding of the survey.

TAKEAWAY 18W

A sightline with no solution is a seat that cannot see, rather than a seat that sees badly.

END OF THE DAY 17W

A ratio falling as the square of a distance, and an equation with two solutions a cycle.

DAY 9 OF 15

Three and a half tonnes, written down

PLAN CARD 99W

Tuesday. The seats arrive at eight and the flying rehearsal is at two. Pell wants the load plot before either. Osei has four pieces to fly. Nobody has tested the grid above them since 1958. Sallow's sequence puts three of the four in the air at once. Every line leans, because the grid is narrower than the stage. Today you work out the tension in each line. Then you total the load on the grid and say what the margin is. A grid signed off on somebody's word is the one number here nobody gets a second chance at.

BRIEFING 15W

Osei will not fly the sequence without it and Pell will not sign without it.

WHAT YOU DECIDE 10W

Produce a load plot the grid can be signed against.

9.1 The one that leans furthest

FLY FLOOR · A A PLACE · BALLPARK

SCENE 35W

The widest piece is a 260 kilogram truss cloth. The grid is narrow enough here that each of its two lines leans 41 degrees off the vertical. The lines are rated at 180 kilograms each.

GUIDE 49W

Work out what each line actually pulls at this angle, then compare it with the rating. Forty-one degrees is a long way from vertical and the cosine is what carries the load, so the penalty is not a few per cent. Say whether the piece can fly as drawn.

ASK 9W

Estimate the tension in each line at 41 degrees.

BEHIND THE BUTTON 99W

Why the penalty accelerates. The cosine falls slowly at first and then quickly: ten degrees costs two per cent, twenty-two costs eight, forty-one costs thirty-three, and sixty doubles the tension. A drawing that leans lines a little is fine and one that leans them a lot is a different order of problem.

What can be done about it. A bridle — a short line pulling the two mains closer to vertical. Or a narrower piece, or the piece flown from a different pair of lines further out. All three are available today, and all three need this number first.

TAKES AS READ

the two lines are symmetric and the piece hangs level · vectors: components, magnitude and direction — taken as read

VERDICT 115W

Two lines at 41 degrees have a combined vertical capability of twice the cosine, which is 1.509. So each pulls 172 kilograms rather than the 130 that halving suggests. Against a 180 kilogram rating that is inside the limit, and only just: four per cent. Nothing is left for the snatch load when a piece is started or stopped quickly. That is the argument for a bridle rather than for a refusal, and it is only visible because the angle was carried through. At 60 degrees the same cloth would put 260 kilograms in each line, which is well past the rating. So the shape of the cosine matters as much as this one number.

TAKEAWAY 18W

The cosine of the lean is what carries the weight, and it falls away fast past thirty degrees.

9.2 A bridle, or a narrower piece

SCENE SHOP · A PERSON · PROTOCOL

SCENE 37W

Three options are on Novak's bench. A bridle that brings each line to 28 degrees, a piece cut 900 millimetres narrower, or the cloth flown from the next pair of lines out. Shop time is four hours.

GUIDE 41W

Each option changes a different quantity in the same relationship. Work out which one changes the angle, which changes the weight and which changes neither. Then rank them by what four hours of shop time can actually deliver before two o'clock.

ASK 10W

Match each option to what it changes in the relationship.

BEHIND THE BUTTON 87W

What a bridle does. It pulls the two main lines toward the vertical by adding a short line between them, so the cosine rises and every tension falls. It costs two shackles and an hour, and it adds a point load where the bridle meets the mains.

What cutting the piece does. Nine hundred millimetres off the width takes about 30 kilograms off the weight and brings the lines in a little, so it helps twice. It is also four hours of shop time and a repaint.

TAKES AS READ

the next pair of lines out is free in the sequence · matrices as transformations of the plane — taken as read

VERDICT 113W

Three of the four move a term in the same relationship and one does not. A bridle raises the cosine, which lowers every tension and takes an hour. A narrower cloth reduces the weight and the lean together, and costs four hours the shop does not really have. Lines further out reduce the lean for nothing, if the sequence leaves them free — which is a question for Haigh rather than for Novak. The counterweight is the one that feels relevant and is not. It balances the set so one person can move it. The tension in the lines above the piece is unchanged by what hangs on the other end of them.

TAKEAWAY 11W

One relationship, three quantities, and each option moves a different one.

9.3 What the grid carries at once

THE HOUSE · A A PLACE · BALANCE

SCENE 35W

The sequence has three pieces in the air together: 260, 190 and 410 kilograms, with a bridled truss adding 60 more. Pell wants a total with a stated margin against the grid's assessed 1,600 kilograms.

GUIDE 50W

Add what is actually hanging, then decide what a margin means. The tensions in the lines are larger than the weights, and the grid feels the tensions, so a total built from weights alone understates what the ironwork carries. Read the panel and report the figure the file should say.

ASK 12W

Read what you need and report the load the grid actually carries.

BEHIND THE BUTTON 89W

Why the total is not the sum of the weights. Every leaning line pulls harder than the weight it carries, and the grid feels each line's pull rather than each piece's mass. On this sequence that difference is about 130 kilograms.

What the margin is for. A snatch — a piece started or stopped fast — puts a transient well above the static load. And a grid untested since 1958 has an assessed capacity rather than a measured one. Pell's file wants both numbers and the gap between them.

TAKES AS READ

the assessed grid capacity of 1,600 kilograms is a static figure · right-triangle ratios, and finding an angle from two sides — taken as read

VERDICT 110W

The pieces and the bridle come to 920 kilograms on the tickets. The term that does not announce itself is the lean. Every line pulls harder than the weight it carries, and the grid feels the tensions rather than the masses. That is another 130 kilograms, on no ticket anywhere because it is not a mass. Total about 1,050 against an assessed 1,600, which is a real margin and one that Pell can be shown. The counterweights are the tempting fifth term and belong to the counterweight frame rather than the grid. Reporting 920 would be an honest reading of every document in the building and thirteen per cent light.

TAKEAWAY 16W

A grid carries line tensions rather than piece weights, and they are not the same total.

END OF THE DAY 17W

A leaning line pulls harder than the load it carries, and the angle sets by how much.

The seat map goes to print at four

PLAN CARD 102W

Wednesday. The seat map goes to the printer at four. Sallow is right about one thing today. Twelve seats pulled out in the last fortnight of a rising sales curve are twelve seats that sell to nobody. The run is costed on nine hundred. Kettleby is just as right about his own point. A seat with no line to the mark is a refund and a complaint. Today you put a price on both sides. You also find the number the argument has been missing all week. Two people are right, and only one of the two numbers has been worked out.

BRIEFING 21W

Sallow is right that a held seat sells to nobody, and Kettleby is right that a blind seat is a refund.

WHAT YOU DECIDE 14W

Decide what happens to twelve seats, and price a house that is not full.

10.1 What the curve says the twelve are worth

BOX OFFICE · A A PLACE · BALLPARK

SCENE 40W

Dowd's returns data gives a resale price that rises as the house fills. With 60 seats left it is £41, with 30 it is £82, and with 12 it is £205. The model is a constant over the seats remaining.

GUIDE 54W

A price that doubles as the seats remaining halve is a constant divided by the seats remaining. Find the constant from one of the three points and use it on the twelve. Then say what the model does as the last seat goes, because that behaviour is the whole reason this shape was chosen.

ASK 9W

Estimate the modelled value of the last twelve seats.

BEHIND THE BUTTON 89W

What the asymptote means here. As the seats remaining approach zero the modelled price rises without bound, which is a vertical asymptote at zero. No real seat sells for an unbounded price, so the model is describing scarcity rather than predicting a transaction.

Why the fit is convincing anyway. Sixty seats at £41, thirty at £82 and twelve at £205 all give a constant of about 2,460. Three points agreeing on the constant is what identifies the family, and it is a stronger test than any of them individually.

TAKES AS READ

the resale price data comes from comparable performances · zeros, multiplicity and what a graph does at one — taken as read

VERDICT 96W

Twelve seats at the modelled £205 is £2,460. That equals the fitted constant, and the reason is the shape of the model: price times seats remaining is fixed. So the last twelve are worth what the last thirty were, and what the last sixty were. It is a real result about scarcity, and it is also where the model stops. As the seats remaining approach zero the price rises without bound, and no seat sells for an unbounded amount. The £2,460 is the number for the argument. The asymptote is the caveat that goes with it.

TAKEAWAY 14W

A constant over a shrinking quantity rises without bound, which is a vertical asymptote.

10.2 Withdraw, discount or reblock

THE HOUSE · A PERSON · CHOICE

SCENE 33W

Twelve seats have no line to the mark. Withdrawing them costs about £2,460 across the run. Reblocking the number moves the mark 1.2 metres downstage, which Mbatha says the band can live with.

GUIDE 55W

Both people in the argument are right about their own number. So the answer is not a compromise between them. It is the third option nobody has costed. Read what each course actually does to the twelve seats and to the show, and pick the one that removes the problem rather than paying for it.

ASK 9W

What should happen before the map goes to print?

BEHIND THE BUTTON 91W

Why discounting is the worst of the three. A seat sold cheap to somebody who then cannot see the performer produces a refund and a complaint, so the revenue is notional and the cost is real. Selling a known blind seat is the one option that gets worse after the transaction.

What reblocking costs. Moving the mark 1.2 metres downstage puts the performer nearer the band, which Mbatha has already accepted. It changes two lighting angles, and Adeyemi can replot those in an hour. It costs an hour and no seats.

TAKES AS READ

the mark can move without changing the staging elsewhere · choosing a model, and what a residual says about the choice — taken as read

VERDICT 105W

Sallow is right that twelve seats withdrawn late in a rising curve are twelve seats sold to nobody. Kettleby is right that a seat with no line is a refund. Both numbers are correct, which is why the answer is the option neither of them has costed. Moving the mark 1.2 metres downstage clears the lines. It costs an hour of replotting and no seats, and the musical director has already agreed to it. Withdrawing pays £2,460 to solve a geometry problem. Discounting sells a known defect and turns revenue into refunds and complaints. Printing as it stands is the same thing without the discount.

TAKEAWAY 17W

When two people are each right about their own number, the answer is usually a third option.

10.3 Two angles, an hour

LIGHTING BOARD · A A PLACE · BALLPARK

SCENE 33W

The mark moves 1.2 metres downstage. The front bar is 6.2 metres up and was 9.4 metres out; the side boom is 4.1 metres up and 3.3 metres across. Both need new angles.

GUIDE 49W

Recompute each angle from its own two sides after the move. The front bar's horizontal distance shortens by 1.2 metres and the boom's does not change at all. That is the part worth noticing before any arithmetic: a move downstage is along one unit's axis and across the other's.

ASK 9W

Estimate the new down angle from the front bar.

BEHIND THE BUTTON 97W

Why the boom barely moves. The side boom is out to the side, so a mark moving downstage changes its distance by very little — the two displacements are nearly perpendicular. The front bar is straight out front, so the same 1.2 metres comes almost entirely off its throw.

What the new front angle is. Six point two over 8.2 gives a tangent of 0.756 and an angle of 37 degrees, up from 33. That is closer to what Adeyemi wanted in the first place, which is the sort of accident worth noticing and not relying on.

TAKES AS READ

the mark moves along the centre line · the unit circle, and sine and cosine as coordinates — taken as read

VERDICT 101W

The bar height is unchanged and the horizontal distance is now 8.2 metres, so the tangent is 0.756 and the angle is 37.1 degrees. Using the old 9.4 gives the angle the plot already had, which is the error that leaves a unit aimed at where somebody used to stand. The interesting part is the asymmetry. The same 1.2 metre move takes 1.2 metres off the front bar's throw and almost nothing off the side boom's. The displacement is along one unit's axis and across the other's. So one angle changes by four degrees and the other by less than one.

TAKEAWAY 18W

A displacement along one unit's axis is across another's, and each angle is recomputed from its own sides.

END OF THE DAY 15W

A quantity rising without bound as its input approaches a value has an asymptote there.

DAY 11 OF 15

Three channels and a matrix

PLAN CARD 100W

Thursday. The recording company is in on the fourteenth. So the wash has to read on camera as well as in the room. Adeyemi has three channels and one filter. Ferreira has dimmers that do not do what their numbers say. Sallow wants the look signed off before the dress rehearsal. Today you work out what a mix of light actually delivers. You also work out what a transformation does to a plot. Three channels can each do exactly what they are told and the look can still be wrong. That is a multiplication somebody has treated as an addition.

BRIEFING 14W

The wash reads green on camera and nobody can say which channel is lying.

WHAT YOU DECIDE 13W

Mix a colour that survives the filter, and say what a transformation does.

11.1 Not the numbers on the desk

LIGHTING BOARD · A A PLACE · PROTOCOL

SCENE 38W

The desk shows red at 80, green at 80 and blue at 40. The filter passes 90 per cent of red, 70 of green and 35 of blue. The dimmer curve delivers the square of the fader setting.

GUIDE 47W

Three things happen to each channel in order: the fader, the dimmer curve and the filter. Each one multiplies, so the order does not change the product — but the sizes do. Work out which channel loses most, and say why the room and the camera disagree.

ASK 7W

Match each channel to what actually arrives.

BEHIND THE BUTTON 97W

Why a dimmer curve exists. A fader at half does not deliver half the light. Older dimmers approximate a square law, so 80 per cent delivers about 64 and 40 per cent delivers 16. That is why the blue is so much weaker than the desk suggests.

Why the camera sees it differently from the eye. The eye adapts to the overall colour of a scene and largely discounts it; a camera does not unless it is told to. So a wash that looks neutral in the room reads green on a recording, which is the complaint.

TAKES AS READ

the dimmer curve is the same on all three channels · vectors: components, magnitude and direction — taken as read

VERDICT 116W

Each channel passes through the fader, the dimmer curve and the filter, and every stage multiplies. Eighty on the desk is about 64 through a square-law dimmer, and 90 per cent of that leaves red at 58. The same 64 through a 70 per cent filter leaves green at 45. Forty on the desk is only 16 through the curve, and 35 per cent of that leaves blue at 6. So the wash has almost no blue in it at all, and that is the green cast on camera. The desk reading of 80, 80, 40 looks balanced and is a set of inputs; what the performer is lit by is the product of three multiplications.

TAKEAWAY 13W

A chain of multiplications is a transformation, and every stage scales what arrives.

11.2 Order matters, or it does not

ORCHESTRA PIT AND SOUND · A PERSON · CHOICE

SCENE 30W

Brand has a compressor and an equaliser in the same channel. Mbatha wants the equaliser first; the patch has the compressor first. Both insist their order is the correct one.

GUIDE 50W

Ask of each pair of operations whether swapping them changes the result. Two gains in series commute, because multiplication does. A compressor responds to the level arriving at it, so what comes before it changes what it does. Work out which of the four pairs commute and which do not.

ASK 11W

Which pair genuinely gives a different result in the other order?

BEHIND THE BUTTON 92W

Why gains commute and processing does not. Multiplying by three then by four is the same as four then three. A compressor is not a multiplication. It applies a different gain depending on the level it sees, so changing what arrives changes the gain it picks.

Why this is the same idea as matrices. A transformation of the plane is applied to every point, and two transformations applied in different orders generally give different results. Rotating then stretching is not stretching then rotating, and the reason is the same as the compressor's.

TAKES AS READ

both devices are in the same channel path · vectors: components, magnitude and direction — taken as read

VERDICT 113W

A compressor chooses its gain from the level arriving at it. Boost a frequency before it and it compresses on that frequency. Boost the same frequency after it and the compression is left alone, then the result is lifted. Those are audibly different, and both engineers are describing a real preference rather than a mistake. Everything else on the list is a multiplication. Two gain stages, a gain and a pad, a channel fader and a group fader all commute. The product of two numbers does not depend on their order. That is the distinction worth carrying, and it is exactly why matrices are introduced with the warning that they do not commute.

TAKEAWAY 7W

Multiplications commute and level-dependent operations do not.

11.3 Two functions of one parameter

SCENE SHOP · A A PLACE · CHOICE

SCENE 31W

The bridled truss rises 4.2 metres while the revolve turns it 1.15 radians. Novak needs to know whether it clears the false proscenium during the move, not just at the ends.

GUIDE 53W

The piece has a height that depends on time and a bearing that depends on time, so its position is two functions of one parameter. Say what that means for the question being asked. A clearance at the start and a clearance at the end do not establish a clearance in the middle.

ASK 9W

What does the piece's path require to be checked?

BEHIND THE BUTTON 92W

Why the ends are not enough. Both coordinates change through the move, so the path between the two endpoints is a curve rather than a straight line. A piece can clear at both ends and foul in the middle, which is exactly the case a rehearsal finds and a drawing does not.

What the parameter is. Time, here — but it could be a fraction of the move, and often is, because the flying operator and the revolve are on different clocks. The useful parameterisation is the one both operators can read.

TAKES AS READ

the two movements start and finish together · converting between polar and rectangular — taken as read

VERDICT 102W

Height and bearing both change through the move, so the piece traces a curve and the clearance has to be evaluated along it. Checking the ends is the standard mistake, and it passes a move that fouls in the middle. That is the failure a rehearsal finds with forty-one people watching. The highest point matters if the obstruction is above, and the false proscenium is beside the path rather than over it, so height alone does not settle it. The total angle is a fact about the revolve and says nothing about where the truss is at any moment during the turn.

TAKEAWAY 15W

Two coordinates changing together trace a curve, and the ends say nothing about the middle.

END OF THE DAY 14W

A transformation applied to every point is a matrix, and it does not commute.

Two equations, one stage

PLAN CARD 92W

Friday, and the dress rehearsal is at seven with Pell in the room. The flying cue and the revolve have to arrive together, which is two rates and one moment. The exit timings have to be walked rather than estimated, because a licence signed on an estimate is signed on nothing. Haigh has both on her sheet and neither has a number yet. Today you solve for when two moving things meet, and time an exit by walking it. Two rates and a distance is a system with one answer, or none.

BRIEFING 15W

The dress is at seven and two things have to happen at the same instant.

WHAT YOU DECIDE 14W

Settle the flying cue against the revolve, and the exit timings against the licence.

12.1 When the two rates meet

FLY FLOOR · A A PLACE · BALLPARK

SCENE 38W

The truss flies out at 0.42 metres a second from 2.4 metres. The revolve brings the rostrum's edge toward the same point at 0.31 metres a second from 6.8 metres away. Haigh needs the moment they are level.

GUIDE 52W

Write each position as a function of time and set them equal. That is a pair of linear equations, and where they agree is a single instant — unless the rates are such that they never do. Work out the time, then check whether it falls inside the length of the cue.

ASK 11W

Estimate the time the truss and the rostrum edge are level.

BEHIND THE BUTTON 103W

What no solution looks like here. If both moved at the same rate in the same direction the gap would never close. The two lines would be parallel, and there would be no moment at all. That is the geometric meaning of an inconsistent system, and it is a real possibility in a flying cue rather than a textbook curiosity.

Why the answer has to be inside the cue. A solution at eleven seconds is useless if the cue is eight seconds long — mathematically fine and operationally a collision, because something else happens at eight. The domain is part of the answer.

TAKES AS READ

both movements run at constant speed through the cue · matrices as transformations of the plane — taken as read

VERDICT 119W

The gap is 4.4 metres and the two are closing it at 0.42 plus 0.31, which is 0.73 metres a second, so they are level at six seconds. Adding the rates is the step to get right. They approach each other, so the closing speed is the sum. Subtracting them gives 40 seconds and a cue that appears to have enormous margin. Six seconds inside an eight-second cue leaves two seconds, which is a margin rather than a coincidence, and it is the number Haigh calls the cue against. Had the rates been equal and in the same direction there would have been no solution at all. In a flying cue that means a collision rather than a curiosity.

TAKEAWAY 15W

Setting two linear positions equal gives one instant, or none if the lines are parallel.

12.2 Walked, not estimated

THE HOUSE · A A PLACE · VERIFY

SCENE 30W

Pell needs the house cleared in under 150 seconds. The estimate on file, made by dividing the seat count by a per-second flow figure, says 138. Nobody has walked it.

GUIDE 47W

Predict the time from the flow figure, then walk the exit with the company and measure it. Lock the prediction first. The value of the measurement is the comparison: a walked figure that agrees with an estimate is a different piece of evidence from an estimate alone.

ASK 13W

Predict the clearance time from the flow figure, then walk it and report.

BEHIND THE BUTTON 82W

Why an estimate from a flow rate is optimistic. It assumes a steady flow through every exit from the first second. Real houses have a delay while people stand and gather, and a bottleneck at the narrowest point. Both push the real figure above the estimate.

What the licence actually wants. A measured time from a walked rehearsal, because the file has to show the room rather than the arithmetic. Pell has read enough files to know the difference and says so.

TAKES AS READ

the walked exit uses the exits the licence relies on · a function as a rule, its domain and what the domain rules out — taken as read

VERDICT 101W

The flow figure predicts 138 seconds and the walk measures 161, which is over the licence limit. The gap is the whole reason the walk is required. A flow calculation assumes steady movement through every exit from the first second. A real house spends time standing up, and then queues at its narrowest point. Both effects push the measurement above the estimate, and neither is visible in the arithmetic. What the walk also gives, which no estimate can, is where the queue formed: the stage-left pass door. That turns a failed number into a specific thing to fix before the fourteenth.

TAKEAWAY 19W

A prediction and a measurement are two pieces of evidence, and the second is the one the file needs.

12.3 Where the curve turned

BOX OFFICE · A PERSON · CHOICE

SCENE 22W

Weekly sales now read 41, 66, 188, 402, 511, 566. Sallow reads it as still accelerating. Dowd says it turned last week.

GUIDE 40W

Work with the differences rather than the totals. Sales are still rising, so the totals say nothing about the argument. What settles it is whether the rise itself is getting bigger or smaller, which is the difference of the differences.

ASK 9W

What do the six figures say about the argument?

BEHIND THE BUTTON 80W

What the two claims actually are. Sallow is claiming the second difference is still positive — the increase is increasing. Dowd is claiming it has gone negative. Both agree sales are rising, which is why the totals cannot arbitrate.

Why it matters commercially. If the rise is slowing, the last hundred seats will not sell themselves and the marketing spend has a job to do. If it is still accelerating, the spend is wasted on seats that were going anyway.

TAKES AS READ

all six figures are complete weeks · average rate of change over an interval — taken as read

VERDICT 103W

The weekly rises are 25, 122, 214, 109 and 55. They grew to week four and have fallen away since. So sales are still increasing while the increase is shrinking. That is precisely Dowd's claim, and precisely why the totals cannot settle it. Sales are not falling: every week is larger than the one before, and reading a slowing rise as a decline is the mirror error. Six points are ample to see a turn in the differences; what six points cannot support is a forecast past them. Commercially the answer decides whether the last hundred seats need spending on, and they do.

TAKEAWAY 15W

Whether a rise is speeding up is the difference of the differences, not the total.

END OF THE DAY 12W

Two things moving at different rates meet at one moment, or never.

DAY 13 OF 15

The rig, in the coordinates it was drawn in

BEFORE THE DAY — CANVASS

Ask the front six rows what they can see

Alan Kettleby, the house manager, will not withdraw a seat on an opinion and will not sell one on a guess. Sit somebody in each of the front rows with the marker on stage, ask one question and count the answers — a sightline is a measurement and the room is the only instrument for it.

PLAN CARD 101W

Saturday. The rig is hung, except for the focus. Ferreira has twenty-eight lamps on four bars and a plot written in bearings. One lamp is not where the plot puts it. The focus session found a shadow on the mark that nothing accounts for. Adeyemi wants the fault found before the notes session at six. Today you use the plot's own coordinates to work out which lamp moved. A rig plotted in one coordinate system and hung in another has exactly one kind of error in it. So there is one thing to look for, and twenty-eight lamps to look at.

BRIEFING 12W

Twenty-eight units are hung and one of them is 900 millimetres out.

WHAT YOU DECIDE 14W

Finish the plot, and find the one unit that is in the wrong place.

13.1 Nine hundred millimetres

LIGHTING BOARD · A A PLACE · BALANCE

SCENE 33W

Four units on bar two are plotted at 6.9 metres and bearings of 12, 24, 36 and 48 degrees. Ferreira's hanging sheet lists offsets of 1.43, 2.81, 4.06 and 4.63 metres from centre.

GUIDE 39W

Convert each plotted bearing into an across-stage offset and compare it with the sheet. Three will agree within the tolerance a tape measure gives; one will not. Work out which, and by how much, before anybody climbs a ladder.

ASK 14W

Read what you need and report where the unit should hang, measured from centre.

BEHIND THE BUTTON 95W

Why the fourth is the suspect. Six point nine metres at 48 degrees gives an offset of 5.13 metres, and the sheet says 4.63. That is half a metre out, which on a focused unit is a shadow on the mark rather than a slightly different look.

Why this error is always the same kind. Somebody converted with the wrong function, or read the bearing as an offset directly. Both produce a number that is plausible, on the right bar, in the right units, and wrong — which is why it survives being written down.

TAKES AS READ

all four units are on the same bar at the same distance · polar coordinates, and a beam described as r of theta — taken as read

VERDICT 107W

Six point nine metres at 48 degrees is an offset of 5.13. The sheet's 4.63 is the same distance at 42 degrees, which is the complement. That is what you get measuring the bearing from the proscenium line instead of the centre line. That is the error the focus session found. The term that does not announce itself is the bar. It was hung to the 1911 drawing, so it sits 140 millimetres off the room's own centre line. No paperwork records that, because the paperwork is what is wrong. The yoke is the tempting fourth term and is zero, because the clamp centres on the bar.

TAKEAWAY 15W

A plausible number in the right units is the error that survives being written down.

13.2 Bearings without a reference

FLY FLOOR · A PERSON · CHOICE

SCENE 25W

Adeyemi's plot lists distances and bearings. It does not say what the bearings are measured from, and there are three plausible references in this building.

GUIDE 44W

A polar description needs three things and the plot has two. Say what is missing. Then say why a set of bearings with no stated reference is not merely inconvenient. It is ambiguous by a fixed amount that nobody can detect from the numbers.

ASK 13W

What does the plot have to state before it can be hung from?

BEHIND THE BUTTON 85W

What a polar coordinate system requires. An origin, a reference direction and a sense of rotation. Miss any one of the three and the numbers describe a family of positions rather than a position.

The three plausible references here. The centre line, the proscenium line and the front edge of bar one, which differ by 90 degrees and by 400 millimetres. Under two of the three, every bearing in the plot is wrong by a fixed amount. That is exactly what today's error looked like.

TAKES AS READ

the plot's distances are all measured from one origin · the unit circle, and sine and cosine as coordinates — taken as read

VERDICT 99W

A bearing is an angle from something, and the plot does not say what. The building offers three candidates: the centre line, the proscenium line, and the front edge of bar one. So every number in the plot names three different positions, and nothing in the numbers reveals which was meant. That is what produced today's half-metre error. Bar heights, dimmer numbers and beam angles are all needed to finish the rig, and none of them resolves the ambiguity. A unit hung at the wrong bearing is in the wrong place however well the rest of its paperwork reads.

TAKEAWAY 14W

A polar description needs an origin, a reference direction and a sense of rotation.

13.3 The centre line the room has

SCENE SHOP · A A PLACE · PROTOCOL

SCENE 40W

Quirke is setting out the false proscenium. The 1911 drawing gives a centre line. The 1958 alteration moved the frame. A tape run from the two side walls gives a third line, 140 millimetres from the one on the drawing.

GUIDE 43W

Three candidate centre lines and one piece to build. Decide which line the piece has to be symmetric about, and say what makes that the right answer rather than the average of the three. The audience is the test rather than the drawing.

ASK 12W

Match each candidate centre line to what it is a record of.

BEHIND THE BUTTON 85W

Why the wall measurement wins. The piece is seen against the frame and the walls, by people sitting in the room. Symmetry that reads to an audience is symmetry about what they can see, which is the room, not the drawing.

Why averaging is the wrong instinct. Three measurements of the same thing average usefully; three different things do not. These are a historical record, an altered structure and a current measurement, and the average of them is a line with no referent at all.

TAKES AS READ

the two side walls are the ones visible to the audience · choosing a model, and what a residual says about the choice — taken as read

VERDICT 101W

The piece is looked at by people sitting in the room. So it has to be symmetric about what they can see, and that is the line the walls give today. The 1911 drawing is a record of the building before the change that caused all this. The frame's own symmetry is a record of the 1958 alteration, which is the discrepancy rather than a reference for it. And the average is the instinct worth resisting. Three measurements of one quantity average usefully. These are three different quantities, so their average is a line nothing in the building is symmetric about.

TAKEAWAY 14W

Averaging three measurements of one thing is sensible; averaging three different things is not.

END OF THE DAY 13W

Converting between two descriptions of a point is where a rig's errors live.

DAY 14 OF 15

Everything at once, and one rehearsal left

PLAN CARD 94W

Sunday, and there is one rehearsal left. The pass door bottleneck is still 161 seconds. The flying cue is timed, but it has not been run with the revolve. The wash reads green on camera. The last hundred seats are not selling themselves. Sallow wants the notes closed and Pell wants the file. Today you decide where the last four hours go. You also finish the two numbers the file is still short of. Four hours is a smaller quantity than that list. That is what makes it a decision rather than a plan.

BRIEFING 14W

One rehearsal, four things to fix, and the licence file closes at noon tomorrow.

WHAT YOU DECIDE 13W

Fit the last of the numbers and spend the last of the time.

14.1 Four hours, six jobs

THE HOUSE · A A PLACE · ALLOCATE

SCENE 30W

Four hours of company time before the dress. Six jobs on Haigh's sheet, two of them required by the licence and four of them wanted by somebody with an argument.

GUIDE 46W

Everything on this list is worth doing and the four hours will not cover it. Two of the six are conditions of the licence, so the doors do not open without them. Spend the pool, and know what the plan cannot answer once it is spent.

ASK 18W

Four hours before the dress. Build the plan, and know what it stops you being able to say.

BEHIND THE BUTTON 74W

What the licence actually requires. A walked exit under 150 seconds and a filed load plot. Neither is negotiable and both are measurable, which is what makes them different from the other four items on the sheet.

Why the flying cue has to be run rather than calculated. Six seconds is the computed answer and a rehearsal is the only thing that establishes the operators can hit it, twice, with a company on stage.

TAKES AS READ

the four hours cannot be extended · choosing a model, and what a residual says about the choice — taken as read

VERDICT 122W

The two licence items are not preferences. The doors do not open without an exit time under 150 seconds and a filed load plot, so the exit walk is bought first whatever else waits. The flying cue is the item where computation has run out. Six seconds is the answer and nobody has hit it with a company on stage. A cue nobody has run is a cue that happens for the first time in front of an audience. The notes are protected because forty-one people cannot open on notes they have not heard. What is left buys the re-block or the camera wash. The honest thing is to say which one is not being done, rather than to plan for both.

TAKEAWAY 15W

A pool smaller than the list is a decision about what will not be done.

14.2 The room is different now

ORCHESTRA PIT AND SOUND · A PERSON · BALLPARK

SCENE 35W

With the seats in, the same clap now leaves 22 per cent of its energy after four seconds rather than seven. Brand has to reset the band's level for a room that no longer rings.

GUIDE 50W

Work out what the change in the room does to the level a listener hears, in decibels, and decide which way the band's own level has to move. A room that rings less returns less sound to the seats, so the same playing arrives quieter than it did in rehearsal.

ASK 9W

Estimate the change in level from the room alone.

BEHIND THE BUTTON 95W

Why a deader room needs more level rather than less. Reverberation adds to what a listener receives — the direct sound plus everything the room returns. Take away the returns and the total falls, so a band that was balanced in an empty house is quiet in a full one.

Why nobody trusts their ears for this. The empty house was audibly wrong and the change is a factor rather than an impression. Brand wants the decibel figure so the desk can be moved by a known amount rather than by consensus in a corridor.

TAKES AS READ

the direct sound from the pit is unchanged by the seats · log properties, and why a product becomes a sum — taken as read

VERDICT 103W

The ratio of reverberant energies is 0.072 over 0.22, which is 0.33, and ten times its logarithm is −4.8 decibels. That is the room's own contribution changing, and it is nearly five decibels — a factor of three in intensity, which nobody balances by ear in a corridor. The sign is the part to get right. The seats absorb, the room returns less, and the total at a listener falls — so the band comes up rather than down. Using twenty as the multiplier doubles the figure and would have the desk moved nine and a half decibels, which is a different show.

TAKEAWAY 14W

A room that returns less sound delivers less total level for the same playing.

14.3 Six seconds, twice

FLY FLOOR · A A PLACE · VERIFY

SCENE 28W

The computed moment is six seconds into the cue. Osei and the revolve operator have never run it together. Haigh has an hour and wants two clean runs.

GUIDE 40W

Predict the moment from the two rates, then run the cue and time what actually happens. The computation is not in doubt; what is in doubt is whether two operators on different clocks can arrive at the same instant, twice.

ASK 13W

Predict the moment the two arrive together, then run the cue and report.

BEHIND THE BUTTON 78W

Why two runs rather than one. One run establishes that it can happen. Two establishes that it can be repeated, which is what a cue is. A sequence that worked once is an anecdote and a sequence that worked twice is a cue Haigh will call.

What the measurement is likely to show. Operators anticipate, so the first run usually arrives early and the second, corrected, arrives late. The useful output is the spread rather than either number.

TAKES AS READ

both operators start on the same call · matrices as transformations of the plane — taken as read

VERDICT 97W

The prediction is six seconds and the two runs came in at 5.4 and 6.3, which is the pattern operators produce: the first anticipates and the second over-corrects. Both are inside the eight-second cue, so the sequence stands. The useful output is not either time but the spread: nearly a second of variation. That is what the margin has to absorb, and it is invisible to any calculation. Two runs rather than one is the whole point. One establishes that it can happen and two establishes that it repeats, and a cue is a thing that repeats.

TAKEAWAY 13W

A computed moment and a rehearsed one are two different pieces of evidence.

END OF THE DAY 18W

A finite pool spent on a list longer than itself is a decision about what is not done.

DAY 15 OF 15

The fourteenth

PLAN CARD 108W

Monday the fourteenth. Pell signs the licence at noon on the strength of the file. The doors open at seven. Eight hundred and sixty-one seats are sold. Sallow wants the last hundred released. Kettleby wants the twelve confirmed clear on the moved mark. Osei wants the load plot signed as it stands. Today you close three numbers. You decide what happens to the last thirty-nine seats, and say what the fortnight actually settled. Everything in this building is now one of two things. It is a measurement with a date on it, or it is somebody's memory. The file decides which of the two the next production inherits.

BRIEFING 17W

Pell signs at noon, the house opens at seven, and everything either measures or it does not.

WHAT YOU DECIDE 12W

Close the file, open the doors, and say what the fortnight established.

15.1 What the file says

THE HOUSE · A A PLACE · PROTOCOL

SCENE 30W

Four statements are drafted for the licence file. Pell will read all four and sign or not sign on the strength of them. Two are measurements and two are not.

GUIDE 43W

Read each statement against what was actually done in the fortnight. A measurement carries the conditions it was taken under; an estimate carries the assumptions it was computed from. Both belong in a file and only one of them can be signed against.

ASK 7W

Match each statement to what supports it.

BEHIND THE BUTTON 91W

What the walk established and the calculation did not. A clearance time of 161 seconds from a walked rehearsal with the full company, and where the queue formed. The 138-second estimate assumed a steady flow that no real house produces.

Why the load plot is the strongest item in the file. Every tension in it was computed from a measured angle and a weighed piece, and the total includes the term that appears on no weight ticket. It is the only number in the building with a margin stated beside it.

TAKES AS READ

the file is read as it stands rather than explained in person · choosing a model, and what a residual says about the choice — taken as read

VERDICT 100W

Three of the four carry the conditions they were taken under. A walked clearance with the bottleneck named. A load plot built from weighed pieces and measured angles. And a seat survey done by sitting in the seats with a marker on stage. The fourth is an estimate from a flow rate, and the walk has already measured 23 seconds more than it predicted. All four can go in the file, and only three can be signed against. The honest form is to keep the estimate with the measurement beside it, so the next manager can see which was which.

TAKEAWAY 10W

A measurement carries its conditions; an estimate carries its assumptions.

15.2 The last hundred seats

BOX OFFICE · A PERSON · CHOICE

SCENE 28W

Eight hundred and sixty-one sold, thirty-nine unsold and the twelve now clear on the moved mark. The weekly rise peaked at week four and has been falling since.

GUIDE 41W

Use what the differences said rather than what the total says. A rising total with a falling rise does not fill a house on its own. So decide whether the last thirty-nine sell themselves, and what the answer implies for tonight.

ASK 8W

What should the box office do at noon?

BEHIND THE BUTTON 87W

What the turn means for the last seats. The rise has halved twice, so the remaining thirty-nine will not go the way the first eight hundred did. They need a reason — day-of-performance release, a price, or somebody telling the town the twelve are back on sale.

Why the twelve matter out of proportion. They are the only seats whose availability is news, because they were withdrawn and are now clear. Everything else on the list is a seat that has been on sale for six weeks.

TAKES AS READ

the twelve seats are confirmed clear on the moved mark · average rate of change over an interval — taken as read

VERDICT 108W

The rise has halved twice, so the remaining thirty-nine are not going to sell themselves in a day. A rising total says nothing about the rate, and the rate is what fills the last of a house. The twelve are the only seats whose availability is actually news, having been withdrawn and then cleared. So releasing them is the one action that generates its own demand. Discounting thirty-nine seats on a sold-out-looking night gives away revenue on seats that walk-ups take anyway. Doing nothing reads the total and ignores the rate. Holding all of them for the recording company empties the two rows the camera is pointed at.

TAKEAWAY 15W

A rising total with a falling rise does not finish the job on its own.

15.3 Four numbers that used to be opinions

SCENE SHOP · A A PLACE · SEQUENCE

SCENE 30W

Quirke has the four things that changed this fortnight written on the shop wall. Novak wants them in an order somebody can act on next time the building goes dark.

GUIDE 51W

Look for what the four have in common rather than what each says. In every case a figure taken from somewhere else was replaced by a measurement made here, with a period or a condition attached to it. Put them in the order of how soon the next production needs each.

ASK 11W

Put the four in the order the next production needs them.

BEHIND THE BUTTON 87W

What each one replaced. A centre line from a 1911 drawing. A clearance from a flow rate. A per-line tension from halving a weight, and a beam angle from a bearing with no stated reference. All four were plausible and none was measured.

Why the order is by urgency rather than by importance. The next production inherits the file. What it needs first is what it cannot work without: the room's own centre line, then the load plot, then the seat map, then the rig's reference direction.

TAKES AS READ

the file is inherited by the next production intact · average rate of change over an interval — taken as read

VERDICT 118W

The order is by what stops work rather than by what matters most. Nothing can be set out without the room's own centre line, and everything built to the 1911 line is 140 millimetres wrong. The load plot comes next because nothing flies without it. The seat map is needed before tickets go on sale, which is weeks rather than days. The rig's reference direction is needed when the rig is plotted, which is last — and it is the one that cost half a metre this time. All four are the same move. A figure inherited from a drawing, a rate or a habit, replaced by a measurement of this room with its conditions written beside it.

TAKEAWAY 15W

An inherited figure and a measured one can agree and are not the same evidence.

END OF THE DAY 15W

A measured local number with a stated period is worth more than an inherited figure.

THE CLOSING CARD

The Ellery opened on the fourteenth, ninety minutes late, to eight hundred and sixty-one of nine hundred seats. The grid carries three and a half tonnes on a load plot that is on a wall rather than in somebody's head. The front six rows lost two seats each to a sightline nobody had drawn, and got them back when the mark moved 1.2 metres downstage. The licence was signed on the strength of a walked clearance of 148 seconds, two inside the limit and thirteen better than the first walk, once the stage-left pass door was propped.

What is unfinished is the roof. The rain that came through it on the ninth is still coming through it, and the run pays for a scaffold in March if the houses hold.

You measured the room before anybody argued about it, so the load plot, the beam angles and the seat map are numbers rather than opinions. Nadia Wren flies twice a night on a margin somebody can point at, and Hettie Prosser watched the whole of it from row A seat 12. You are the reason twelve seats that could not see were found in a rehearsal instead of by an audience. Forty-one people worked a run that opened because the arithmetic was finished in time.

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